ERDŐS PROBLEM 878, QUESTION 3B: THE MAXIMA OF f AND F DIFFER INFINITELY OFTEN

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1. The problem

Throughout, p denotes a prime. For an integer $n \geq 1$ let $\mathcal{P}(n)$ be the set of primes dividing n, let $\omega(n) = |\mathcal{P}(n)|$, and let

$$q_p(n) = p^{\lfloor \log_p n \rfloor}$$

be the largest power of p that does not exceed n. Erdős [1] considered the functions

$$f(n) = \sum_{p \in \mathcal{P}(n)} q_p(n) \qquad \text{and} \qquad F(n) = \max_A \sum_{a \in A} a,$$

where the maximum is taken over all sets $A \subseteq \{2, 3, \ldots, n\}$ of pairwise coprime integers all of whose prime factors divide n; we call such sets *admissible for* n. The set $\{q_p(n) : p \in \mathcal{P}(n)\}$ is admissible, so $F(n) \geq f(n)$. For real $x \geq 1$ put

$$M_f(x) = \max_{1 \leq n \leq x} f(n), \qquad M_F(x) = \max_{1 \leq n \leq x} F(n);$$

thus $M_F(x) \geq M_f(x)$, and both functions are non-decreasing. Erdős asked whether $M_f(x) = M_F(x)$ for all sufficiently large x [2]. The answer is no.

Theorem. *For every X_0 there is an integer $x \geq X_0$ with $M_f(x) < M_F(x)$. In particular, $M_f(x) = M_F(x)$ fails for arbitrarily large x.*

Lower bounds for F will always come from the following observation.

Remark (Building admissible sets). Let $a_1, \ldots, a_k \in \{2, \ldots, n\}$, and suppose that their sets of prime factors are pairwise disjoint subsets of $\mathcal{P}(n)$. Then $a_1, \ldots, a_k$ are pairwise coprime, hence distinct, so $\{a_1, \ldots, a_k\}$ is admissible for n and $a_1 + \cdots + a_k \leq F(n)$.

2. Idea of the proof

Suppose that $M_F(x) = M_f(x)$ for all large x, and let $n \leq y$ be a point where f attains its maximum $M_f(y)$ on $[1, y]$. The value $f(n)$ is a sum of prime powers, each at most n, one for each prime factor of n. We construct a number m only slightly larger than n with $F(m) \geq f(n) + Y/25$, where $Y \leq y \leq 1.05\,Y$ for a fixed scale Y:

- m is divisible by almost all prime factors of n, so $F(m)$ can reuse the corresponding terms of $f(n)$;
- two large prime factors r_a, r_b of n are dropped, which loses at most $2n$;
- six fixed primes $s_1, \ldots, s_6$ divide m. The scale Y is chosen so that their terms in $f(n)$ are small (less than $\frac{6}{7}n$ in total), and in $F(m)$ we replace them by three numbers $b_1, b_2, b_3 \in (0.98\,m, m]$, each built from two of the s_i, which contribute more than $2.94\,m$.

Since $2.94 > 2 + \frac{6}{7}$, this gains a fixed proportion of n. As $M_F = M_f$, the function M_f must therefore increase by at least $Y/25$ on the short interval from y to $y + (m - n)$. If $r_a, r_b \geq M$, the step $m - n$ is at most $2CY/M^2$, where $C = s_1 \cdots s_6$; and n has more than M prime factors as soon as $M_f(y)/y$ is large compared with M. Iterating, M_f grows faster and faster while the steps shrink like $1/M^2$, so infinitely many jumps fit into the bounded window $[Y, 1.05\,Y]$, which is impossible.

3. Four lemmas

Lemma 1 (Smooth numbers just below n)**.** *Let $p < r$ be primes with $98r < 100p$. For every $n \geq p^{p^2+2}$ there is an integer z with $0.98\,n < z \leq n$ all of whose prime factors lie in $\{p, r\}$; moreover $z \geq 2$.*

Proof. Let $\ell = \lfloor \log_p n \rfloor$, so that $p^\ell \leq n < p^{\ell+1}$ and $\ell \geq p^2 + 2$. Put $J = p^2 + 1 < \ell$ and

$$z_j = p^{\ell-j} r^j \qquad (0 \leq j \leq J).$$

Each term of this sequence is $r/p < 100/98$ times the previous one. It starts at $z_0 = p^\ell \leq n$ and ends above n: since $r \geq p + 1$,

$$z_J \geq p^{\ell-J}(p+1)^J \geq p^{\ell-J}\big(p^J + Jp^{J-1}\big) = p^\ell + p^{\ell+1} + p^{\ell-1} > p^{\ell+1} > n.$$

Let $j < J$ be the last index with $z_j \leq n$. Then $n < z_{j+1} = z_j \cdot r/p < \frac{100}{98} z_j$, so $z = z_j$ satisfies $0.98\,n < z \leq n$. Also $z \geq z_0 \geq p \geq 2$. □

Lemma 2 (Maximizers lie in the upper half)**.** *For every $n \geq 1$ we have $f(2n) > f(n)$. Consequently, if $1 \leq n \leq y$ and $f(n) = M_f(y)$, then $n > y/2$.*

Proof. Every prime factor p of n divides $2n$, and $q_p(2n) \geq q_p(n)$. The term for $p = 2$ strictly increases: if $2 \mid n$, then $q_2(2n) = 2q_2(n)$, and if $2 \nmid n$, then $f(2n)$ has the additional term $q_2(2n) \geq 2$. Hence $f(2n) > f(n)$. If $n \leq y/2$, then $2n \leq y$ and $f(2n) > f(n) = M_f(y)$, which is impossible. □

Lemma 3 (M_f grows faster than linearly)**.** *For every $\lambda > 0$ we have $M_f(y) > \lambda y$ for all sufficiently large y.*

Proof. Since $\sum_p 1/p$ diverges, there is N with $\sum_{p<N} 1/p \geq 2\lambda + 1$. Let $P = \prod_{p<N} p$ and $y \geq 2P$. The number $t = P\lfloor y/P \rfloor$ satisfies $y/2 < t \leq y$ (because $t > y - P \geq y/2$) and is divisible by every prime $p < N$, and $q_p(t) > t/p$ for each such p. Hence

$$M_f(y) \geq f(t) \geq \sum_{p<N} q_p(t) > t \sum_{p<N} \frac{1}{p} \geq (2\lambda+1)\,t > (2\lambda+1)\frac{y}{2} > \lambda y. \qquad \square$$

Six special primes. Let

$$s_1 = 168000061, \quad s_2 = 168000101, \quad s_3 = 168000121,$$
$$s_4 = 168000139, \quad s_5 = 168000143, \quad s_6 = 168000167.$$

These six numbers are primes with $s_1 < s_2 < \cdots < s_6$, and apart from this the proof uses only the following two properties:

- $98\,s_{2i} < 100\,s_{2i-1}$ for $i = 1, 2, 3$ (the primes differ by less than 200);
- every s_i is larger than $14 \cdot \frac{21}{20} \cdot 15^6 = 167\,442\,187.5$.

Let $S = \{s_1, \ldots, s_6\}$ and $C = s_1 s_2 \cdots s_6$.

Lemma 4 (Choosing the scale)**.** *Call an integer $Y \geq 1$* good *if $q_p(\lfloor \frac{21}{20} Y \rfloor) < \frac{1}{14} Y$ for every $p \in S$. For every integer $Y_0 \geq 1$, one of the seven numbers $Y_0, 15Y_0, 15^2Y_0, \ldots, 15^6 Y_0$ is good.*

Proof. Fix $p \in S$; we show that the inequality for p fails for at most one of the numbers $Y_i = 15^i Y_0$. Suppose it fails for Y_i and Y_j with $i < j$, and let $u = q_p(\lfloor \frac{21}{20} Y_i \rfloor)$ and $v = q_p(\lfloor \frac{21}{20} Y_j \rfloor)$, so that $u \geq \frac{1}{14} Y_i$ and $v \geq \frac{1}{14} Y_j$. Since $Y_j \geq 15\,Y_i$,

$$u \leq \tfrac{21}{20} Y_i < \tfrac{15}{14} Y_i \leq \tfrac{1}{14} Y_j \leq v.$$

Thus $v > u$, and as both are powers of p, $v \geq p\,u \geq \frac{p}{14} Y_i$. On the other hand, $v \leq \frac{21}{20} Y_j \leq \frac{21}{20} \cdot 15^6\, Y_i$. Together these give $p \leq 14 \cdot \frac{21}{20} \cdot 15^6$, contrary to the choice of the s_i. Since S has six elements and there are seven numbers Y_i, some Y_i satisfies the inequality for every $p \in S$. □

4. The key construction

Let $N_0 = s_5^{s_5^2+2}$. Since $s_1 < s_3 < s_5$, Lemma 1 applies to each of the pairs (s_1, s_2), (s_3, s_4), (s_5, s_6) for every $n \geq N_0$.

Lemma 5 (Three pairs)**.** *Let $m \geq N_0$ be divisible by C, and let R be a set of primes dividing m with $R \cap S = \emptyset$. Then*

$$F(m) > 2.94\, m + \sum_{p \in R} q_p(m).$$

Proof. By Lemma 1, for $i = 1, 2, 3$ there is an integer b_i with $0.98\, m < b_i \leq m$ whose prime factors lie in $\{s_{2i-1}, s_{2i}\}$. The numbers b_1, b_2, b_3 and $q_p(m)$ for $p \in R$ lie in $\{2, \ldots, m\}$, and their sets of prime factors are contained in $\{s_1, s_2\}$, $\{s_3, s_4\}$, $\{s_5, s_6\}$, and $\{p\}$ for $p \in R$, respectively. These are pairwise disjoint subsets of $\mathcal{P}(m)$, so by the Remark

$$F(m) \geq b_1 + b_2 + b_3 + \sum_{p \in R} q_p(m) > 3 \cdot 0.98\, m + \sum_{p \in R} q_p(m). \qquad \square$$

5. Proof of the theorem

Assume, for a contradiction, that there is X_0 with

$$M_F(x) \leq M_f(x) \qquad \text{for all integers } x \geq X_0. \tag{1}$$

Put $B = 2000\, C + 1$. Fix a good integer Y such that

$$Y \geq X_0, \qquad Y \geq 2N_0, \qquad M_f(Y) > 2(B + 10)\, Y; \tag{2}$$

such Y exists by Lemma 4 applied to an integer Y_0 larger than X_0, $2N_0$, and the threshold given by Lemma 3 for $\lambda = 2(B + 10)$.

Lemma 6 (One jump)**.** *Let $M \geq 1$ and y be integers with $Y \leq y \leq \frac{21}{20} Y$ and $M_f(y) \geq \frac{21}{20}(M + 8)\, Y$. Then there is an integer y' with*

$$y \leq y' \leq y + \frac{2CY}{M^2} \qquad \text{and} \qquad M_f(y') > M_f(y) + \frac{Y}{25}.$$

Proof. Choose $n \leq y$ with $f(n) = M_f(y)$. By Lemma 2, $n > y/2 \geq Y/2$; hence $Y < 2n$ and, by (2), $n > N_0$. Also $n \leq \lfloor \frac{21}{20} Y \rfloor \leq 2Y$.

Many prime factors. Every term of $f(n)$ is at most n, so

$$\tfrac{21}{20}(M + 8)\, Y \leq f(n) \leq \omega(n)\, n \leq \tfrac{21}{20}\, \omega(n)\, Y,$$

and therefore $\omega(n) \geq M + 8$.

Two large primes. At most 6 prime factors of n lie in S, and fewer than M primes are smaller than M. Hence n has two distinct prime factors $r_a, r_b \notin S$ with $r_a \geq M$ and $r_b \geq M$.

The number m. Let $R = \mathcal{P}(n) \setminus (S \cup \{r_a, r_b\})$ be the set of the remaining prime factors of n, and let $D = C \prod_{p \in R} p$. Since $r_a r_b \prod_{p \in R} p$ divides n,

$$\prod_{p \in R} p \leq \frac{n}{r_a r_b} \leq \frac{2Y}{M^2}, \qquad \text{so} \qquad D \leq \frac{2CY}{M^2}.$$

Let m be the smallest multiple of D that is larger than n. Then $n < m \leq n + D$, and every prime in $R \cup S$ divides m.

Upper bound for $f(n)$. The prime factors of n are the primes in R, the primes r_a and r_b, and possibly some primes in S. For $p \in S$, since Y is good and $n \leq \lfloor \frac{21}{20} Y \rfloor$, we have $q_p(n) \leq q_p(\lfloor \frac{21}{20} Y \rfloor) < \frac{Y}{14} < \frac{n}{7}$. For $p \in \{r_a, r_b\}$ we use $q_p(n) \leq n$, and for $p \in R$ we use $q_p(n) \leq q_p(m)$. Hence

$$f(n) < \frac{6}{7}\, n + 2n + \sum_{p \in R} q_p(m) = \frac{20}{7}\, n + \sum_{p \in R} q_p(m).$$

Lower bound for $F(m)$. As $m > n > N_0$, Lemma 5 applies to m and R and gives

$$F(m) > 2.94\,m + \sum_{p\in R} q_p(m) > 2.94\,n + \sum_{p\in R} q_p(m).$$

Comparison. Subtracting the two bounds and using $n > Y/2$,

$$F(m) - f(n) > \Big(2.94 - \frac{20}{7}\Big)n = \frac{29}{350}\,n > \frac{29}{700}\,Y > \frac{Y}{25}.$$

Now let $y' = y + D$. Then $y \le y' \le y + \frac{2CY}{M^2}$, $m \le y'$, and $y' \ge Y \ge X_0$, so by (1)

$$M_f(y') \ge M_F(y') \ge F(m) > f(n) + \frac{Y}{25} = M_f(y) + \frac{Y}{25}. \qquad \square$$

Iterating the jump. We construct integers $y_0 \le y_1 \le y_2 \le \cdots$ such that for every $k \ge 0$

$$Y \le y_k \le Y + 54\,C\,Y\Big(\frac{1}{B} - \frac{1}{B+k}\Big) \qquad \text{and} \qquad M_f(y_k) \ge \frac{21}{20}(B+9+k)\,Y. \tag{3}$$

Since $B > 1080\,C$, we have $54\,C\,Y/B < Y/20$, so all points considered below lie in the window $[Y, \frac{21}{20}Y]$.

For $k = 0$ take $y_0 = Y$; the second condition holds because $M_f(Y) > 2(B+10)\,Y \ge \frac{21}{20}(B+9)\,Y$.

Suppose y_k satisfies (3). Put $M = B + 1 + k$, so that $\frac{21}{20}(M+8)\,Y = \frac{21}{20}(B+9+k)\,Y$, and apply Lemma 6 27 times in succession, starting from y_k; let y_{k+1} be the last point obtained. Each application is legitimate. Indeed, M_f is non-decreasing and the points increase, so the condition $M_f(y) \ge \frac{21}{20}(M+8)\,Y$ persists. Moreover, after $j \le 27$ jumps the point is at most $y_k + j \cdot 2CY/M^2$, so the total distance travelled is at most

$$27 \cdot \frac{2CY}{(B+1+k)^2} \le \frac{54\,C\,Y}{(B+k)(B+k+1)} = 54\,C\,Y\Big(\frac{1}{B+k} - \frac{1}{B+k+1}\Big),$$

so every point stays at most $Y + 54\,C\,Y\big(\frac{1}{B} - \frac{1}{B+k+1}\big) < \frac{21}{20}Y$. In particular, y_{k+1} satisfies the first condition of (3). Since each jump increases M_f by more than $Y/25$,

$$M_f(y_{k+1}) > M_f(y_k) + \frac{27}{25}\,Y \ge \frac{21}{20}(B+9+k)\,Y + \frac{21}{20}\,Y = \frac{21}{20}(B+10+k)\,Y,$$

which is the second condition for $k+1$.

The contradiction. By (3), every y_k is at most $\frac{21}{20}Y$, so $M_f(y_k) \le M_f(\frac{21}{20}Y)$, a number that does not depend on k. On the other hand, $M_f(y_k) \ge \frac{21}{20}(B+9+k)\,Y \ge k$ for every k. Taking $k > M_f(\frac{21}{20}Y)$ gives a contradiction, which proves the Theorem. $\square$

Remark (How the constants fit together). The numbers in the proof are tied together as follows.

- The gain in one jump comes from $3 \cdot 0.98 = 2.94 > 2 + \frac{6}{7}$, and $\big(2.94 - \frac{20}{7}\big) \cdot \frac{1}{2} = \frac{29}{700} > \frac{1}{25}$.
- 27 jumps raise M_f by more than $\frac{27}{25}Y > \frac{21}{20}Y$, which is enough to raise M by one.
- The steps have length at most $2CY/M^2$ and $\sum_M 1/M^2$ converges. Starting from $M = B + 1$ with $B > 1080\,C$, the total distance is at most $54\,C\,Y/B < Y/20$, so we never leave the window $[Y, \frac{21}{20}Y]$, on which the goodness of Y controls the terms of the six special primes.
- The bound $s_i > 14 \cdot \frac{21}{20} \cdot 15^6$ makes the pigeonhole argument of Lemma 4 work, and $98\,s_{2i} < 100\,s_{2i-1}$ is what Lemma 1 needs.

The same argument is formalized in Lean in `FormalConjectures/ErdosProblems/878.lean` (theorem `erdos_878_q3_disproved`).

References

[1] P. Erdős, *On two unconventional number theoretic functions and on some related problems*, 1984.
[2] *Erdős Problem #878*, `https://www.erdosproblems.com/878`.